

TRUSTIA AI — ALCHEMIST ACCELERATOR DOSSIER

OFFICIAL APPLICATION DOSSIER // #1 ENTERPRISE & DEEP-TECH ACCELERATOR (\$125,000 INVESTMENT)

INVESTMENT ASK: \$125K SAFE \$5M Post-Money Cap	CODE & TESTS: 1,281 TESTS (100%) 16K Lines Deterministic	DEFENSE RATING: 100/100 PERFECT SSB & BTK Verified	INCUBATION: ITO BTM & TEKNOPARK Polat Tower & Cube GO
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1. Elevator Pitch & Problem Statement

TRUSTIA AI is a deterministic, hardware-agnostic Level 4 autonomy software stack that converts military ground vehicles (UGVs) and commercial platforms into fully autonomous systems in 15 minutes, operating with sub-8cm precision in electronic warfare and GPS-denied combat environments.

The Problem: Modern defense operations face intense GPS jamming and asymmetric hazards. Commercial robotaxi stacks fail when satellite GNSS is severed. Furthermore, defense OEMs spend \$5M–\$15M and 2–3 years developing custom autonomy software for every mechanical vehicle chassis. Trustia solves this with a plug-and-play universal software "brain" integrating a 400Hz ESKF filter, 3D LiDAR SLAM, and Hybrid A* planning.

2. Deep-Tech Moat & Technological Differentiation

400Hz GPS-Denied ESKF	Fuses 3D LiDAR + 400Hz IMU. Position drift <0.08m over 10km in electronic warfare theaters.	Anti-Jamming / Spoofing
15-Minute Plug & Play	Direct CAN 2.0B / CAN FD / SAE J1939 Drive-by-Wire interface. Zero mechanical redesign needed.	SAE J1939 / ISO 26262
1,281 Automated Tests	Zero external critical dependencies. Deterministic real-time cycle latency < 1.2 ms.	1,281 / 1,281 (100% PASS)
Multimodal Threat Isolation	LWIR Thermal (FLIR Boson) + Induction + GPR Radar fusion to isolate explosive hazards autonomously.	NATO STANAG 4586

3. Market Size (TAM / SAM / SOM) & Enterprise Unit Economics

- TAM (\$34.8B): Global military Unmanned Ground Vehicle (UGV) market + \$120B autonomous commercial mobility market by 2032.
- SAM (\$6.2B): NATO & allied nation autonomous ground vehicle retrofit and software modernization market.
- SOM (\$180M): Initial capture across 1,200 vehicles in Turkey, US, and Europe over 5 years.

Total Revenue (\$)	\$240,000	\$1,150,000	\$3,800,000
Hardware Kits Deployed (\$80k-\$220k)	6 Vehicles (PoC)	35 Platforms	140+ Platforms
Annual C2 / SaaS Licenses (\$15k-\$35k)	\$90,000	\$525,000	\$2,100,000
Gross Margin (%)	84%	87%	91%
Team Size (Engineers & Operations)	4 Team Members	9 Team Members	20 Team Members

4. Founder Solvency & Government Accreditations

- Founder & Chief Executive Officer (CEO): Murat Furkan Bayram.
- Incubator Backing: Admitted to Istanbul Chamber of Commerce BTM Incubator (Polat Tower, Istanbul).
- Defense Tech Hub: Candidate technology firm in Teknopark Istanbul (Cube Incubation / Cube GO).
- Defense Industry Excellence Rating: 100/100 Perfect Grade by Turkish Presidency of Defense Industries (SSB & BTK, ID: L2zPtN4X1ZJ).
- Certified Advanced Entrepreneur: Accredited by the Ministry of Industry and Technology (KOSGEB KSB01UGE0115153370).
- National R&D; Registry: Registered in TÜBİTAK ARBİS (TBTK-0229-6571). Active ASELSAN Defense Supplier Portal Registration.

5. Why Alchemist & Submission Next Steps

WHY ALCHEMIST: Expand into US defense tech & enterprise dual-use mobility markets, incorporate Delaware C-Corp, and connect with Tier-1 enterprise partners and Silicon Valley Deep-Tech venture funds.
PORTAL SUBMISSION: Direct submission at alchemistaccelerator.com/apply (Questions 1–8 attached).